

Munkres 11

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1 Free Products

Let $\{G_\alpha\}$ be a collection of subgroups of G such that if $\alpha \neq \beta$, $G_\alpha \cap G_\beta = 1$. Suppose further that any $x \in G$ has a unique reduced word representation (this implies G is generated by $\{G_\alpha\}$). Then G is said to be the free product of $\{G_\alpha\}$.

Let $h_\alpha : G_\alpha \rightarrow K$ be homomorphisms. Then the $\{h_\alpha\}$ extend uniquely to $h : G \rightarrow H$.

Let W denote the set of all words in $\{G_\alpha\}$. Let $V : W \rightarrow G$ be the valuation map, $H : W \rightarrow K$ be the product of component-wise application of h_{α_i} . For words w, w' let ww' denote concatenation. Note $V(ww') = V(w)V(w')$ and $H(ww') = H(w)H(w')$, so V, H preserve “multiplication.” For $x \in G$, let $R(x)$ denote the reduced word representing x . Define $h : G \rightarrow H$ by $h(x) = H(R(x))$. I claim h is the homomorphism in question.

Suppose $V(w) = x$. I claim $H(w) = h(x)$. Let w' be a reduction of w . Then $V(w') = V(w) = x = V(R(x))$. Since $R(x), w'$ are reduced, $R(x) = w'$. Then $H(w) = H(w') = H(R(x)) = h(x)$, as desired.

Now: $V(R(x)R(y)) = V(R(x))V(R(y)) = xy$, so $H(R(x)R(y)) = h(xy) \rightarrow h(xy) = H(R(x))H(R(y)) = h(x)h(y)$. If $x \neq 1 \in G_\alpha$, then $R(x) = (x)$, so $h(x) = H((x)) = h_\alpha(x)$. Else, $x = 1$ and $h(x) = H(\emptyset) = 1$.

Suppose h' is another homomorphism satisfying the given properties. Then

$$h(x) = h\left(\prod x_i\right) = \prod h(x_i) = \prod h_{\alpha_i}(x_i) = \prod h'(x_i) = h'\left(\prod x_i\right) = h'(x).$$

2 Free Groups

Let $G = \langle g_1, g_2, \dots, g_n \rangle$ be a finitely generated free group. Let the associated monomorphisms be $i_\alpha : \mathbb{Z} \rightarrow G$. g_i for $i \in [n]$ is a finite product of elements of the form $i_\alpha(z)$. Let J_i denote the set of occurrences of α . Let $J_{[n]} = \bigcup_{i \in [n]} J_i$, so $J_{[n]}$ is finite. Suppose FTSOC J is infinite. Pick $\alpha \notin J_{[n]}$. Since the g_i

generate G , $i_\alpha(1) = \prod g_i$. However, the terms on the RHS only involve indices in $J_{[n]}$, so the LHS and RHS have distinct reduced representations, absurd.

Suppose G is free finitely generated. Let $\{i_k\}, \{j_k\}$ be the generating monomorphisms $\mathbb{Z} \rightarrow G$, where the first k ranges over $[n]$, the second, $[m]$. Let $\mathcal{H}$ denote the set of homomorphisms $G \rightarrow C_2$ satisfying the extension property. Then $C_2^n \longleftrightarrow \mathcal{H} \longleftrightarrow C_2^m$, so $2^n = 2^m \rightarrow n = m$. Hence, we can define the rank r of a free finitely generated group.

Let $h_1 : G_1/[G_1, G_1] \rightarrow H, h_2 : G_2/[G_2, G_2] \rightarrow H$ be homomorphisms into the abelian group H . Let $p_1 : G_1 \rightarrow G_1/[G_1, G_1], p_2 : G_2 \rightarrow G_2/[G_2, G_2]$ be the projection maps. Then $h_1 \circ p_1 : G_1 \rightarrow H, h_2 \circ p_2 : G_2 \rightarrow H$ are homomorphisms. By extension property of G , let $k : G \rightarrow H$ be the extension. Since H is Abelian, k descends to a homomorphism $h : G/[G, G] \rightarrow H$. I claim h extends h_1, h_2 by the homomorphisms $i_1 : G_1/[G_1, G_1] \rightarrow G/[G, G], i_2 : G_2/[G_2, G_2] \rightarrow G/[G, G]$.

$$h \circ i_1(g_1[G_1, G_1]) = h(g_1[G, G]) = k(g_1) = h_1(p_1(g_1)) = h_1(g_1[G_1, G_1]),$$

so $h \circ i_1 = h_1$. Similarly, $h \circ i_2 = h_2$. Hence, by extension property, $G/[G, G] \cong (G_1/[G_1, G_1]) \oplus (G_2/[G_2, G_2])$ by the monomorphisms i_1, i_2 .

Assume the hypotheses of SVK. Let N be the normal subgroup of $G_1 * G_2$ generated by words of the form $(\iota_1(i_1(g)), \iota_2(i_2(g)^{-1}))$, where $g \in \pi_1(U \cap V, x_0)$ and ι_1, ι_2 are usual monomorphisms of G_1, G_2 into $G_1 * G_2$. Let $H = (G_1 * G_2)/N$. Then $H \cong \pi_1(X, x_0)$.

(1) Let $j : G_1 * G_2 \rightarrow \pi_1(X, x_0)$ be the extension of j_1, j_2 . Observe that for any $\gamma \in \pi_1(X, x_0)$, we may write $\gamma = \prod j_{a_i}(\gamma_i) = \prod j \iota_{a_i}(\gamma_i) = j(\prod \iota_{a_i}(\gamma_i))$, so that j is an epimorphism.

Note that

$$\begin{aligned} j(\iota_1(i_1(g))\iota_2(i_2(g)^{-1})) &= j(\iota_1(i_1(g)))j(\iota_2(i_2(g)^{-1})) = j_1(i_1(g))j_2(i_2(g)^{-1}) \\ &= (j_1 i_1)(g)(j_1 i_1)(g)^{-1} = 1. \end{aligned}$$

Thus, $N < \ker j$. Let j descend to an epimorphism $k : H \rightarrow \pi_1(X, x_0)$.

(2) Let $\phi_1 : \pi_1(U, x_0) \rightarrow H$ be $\phi_1 = p \circ \iota_1$, where $p : G_1 * G_2 \rightarrow H$ is the projection onto N . Let ϕ_2 be defined similarly. Note for $g \in \pi_1(U \cap V, x_0)$,

$$\phi_1 i_1(g) = p \iota_1 i_1(g) = \iota_1 i_1(g)N = \iota_2 i_2(g)N,$$

so $\phi_1 \circ i_1 = \phi_2 \circ i_2$. By SVK, there exists $\phi : \pi_1(X, x_0) \rightarrow H$ such that $\phi \circ j_1 = \phi_1, \phi \circ j_2 = \phi_2$.

I claim ϕ is a left inverse of k , which implies k is an isomorphism. It suffices to show $\phi \circ k$ fixes the generators of H . Suppose $g \in \pi_1(U, x_0)$.

$$\phi k(\iota_1(g)N) = \phi j \iota_1(g) = \phi j_1(g) = \phi_1(g) = p \circ \iota_1(g) = \iota_1(g)N.$$

3 Group Theory

Let G be a group and $N \triangleleft G$ be normal. Suppose $H < G/N$. There exists $K < G$ such that $K/N = H$.

Let $K = \bigcup H$. Since $N \in H$, $1 \in K$, so $K \neq \emptyset$. I claim $k \in K$ iff $kN \in H$. Suppose $k \in K$. Then $k \in \bigcup H$, so $k \in gN$ for some $gN \in H$. Then $kN = gN \in H$. Conversely, suppose $kN \in H$. Then $k \in kN \subset K$, so $k \in K$.

Suppose $k, k' \in K$. Then $kN, k'N \in H$. Since H is closed under $\cdot$, $kk'N = (kN)(k'N) \in H$, so $kk' \in K$. Notice that $N \subset K$. Now for inverses. Suppose $kN \in H$. Since H is closed under inverses, there exists $k'N \in H$ such that $(kN)(k'N) = N \rightarrow kk' \in N$. Write $kk' = n$ for some $n \in N$, so $k(k'n^{-1}) = 1 \rightarrow k^{-1} = k'n^{-1}$. Since $k', n^{-1} \in K$, $k^{-1} \in K$ (by closure under $\cdot$). Hence, $K < G$.

It remains to show $K/N = H$. Since $N \triangleleft G$ and $K < G$, we have $N \triangleleft K$, so K/N has a group structure. Further, if $k \in K$, $kN \in K/N$, and $kN \in H$, so $K/N \subset H$. Conversely, suppose $kN \in H$. Then $k \in K$, so $kN \in K/N$. Hence, $H = K/N$, as desired.

Third Isomorphism theorem. Suppose $N < H < G$ and $N \triangleleft G$ (so $N \triangleleft H$ as well). Then $(G/N)/(H/N) \cong G/H$ (it is implicit $H/N \triangleleft G/N$).

Suppose for groups H, G such that $N \triangleleft H, G$ that $H/N < G/N$. Then $H < G$. The proof is easy: $H = \bigcup (H/N) \subset \bigcup (G/N) = G$, so $H \subset G$, but H, G are groups, so $H < G$. We can sharpen to $H/N \triangleleft G/N$.

Let G be a group and $\emptyset \subsetneq A \subset G$. Then denote by $N(A)$ the minimal normal subgroup of G containing A . Note if $A \subset B$, then $A \subset B \subset N(B)$, so $N(B)$ is normal containing A , and therefore $N(A) \subset N(B)$. Suppose $A, B \neq \emptyset$ such that $A \subset N(B)$, $B \subset N(A)$. Then $N(A) \subset N(N(B)) = N(B)$ and similarly, $N(B) \subset N(A)$, so $N(A) = N(B)$.

Now the result. Let $M = N(T \cdot N(R))$ where $T \cdot N(R)$ is a finite set of generators. Assume that T is closed under inverses. I claim $N(T \cdot N(R)) = N(T \cup R)/N(R)$. Suppose $gxg^{-1}N(R) \in N(T \cup R)/N(R)$ for $x \in T \cup R \cup R^{-1}$. If $x \in R$, we're done. Else, assume $x \in T$, so $gxg^{-1}N(R) = (gN(R))(xN(R))(gN(R))^{-1} \in N(T \cdot N(R))$. The converse holds.

Note the third isomorphism theorem applies to $N(R) < N(T \cup R) < F(S)$.

Now:

$$\begin{aligned}
G/M &= \langle S \mid R \rangle / N(T \cdot N(R)) \\
&= (F(S)/N(R)) / (N(T \cup R)/N(R)) \\
&= F(S)/N(T \cup R) \\
&= \langle S \mid T \cup R \rangle,
\end{aligned}$$

which is finitely presented since $S, T \cup R$ are finite.

4 Disjoint Union and Wedge Product

Let X, Y be disjoint topological spaces. Give $X \sqcup Y$ the topology $\mathcal{T} = \{U \cup V : U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$. Indeed, $\emptyset, X \cup Y \in \mathcal{T}$, for arbitrary union $\bigcup_{\alpha \in J} (U_\alpha \cup V_\alpha) = \bigcup_{\alpha \in J} U_\alpha \cup \bigcup_{\alpha \in J} V_\alpha$, and for finite intersection, $\bigcap_{i \in [n]} (U_i \cup V_i) = \bigcap_{i \in [n]} U_i \cup \bigcap_{i \in [n]} V_i$, so $\mathcal{T}$ is indeed a topology on $X \sqcup Y$.

Notice that inclusion $i_x : X \rightarrow X \sqcup Y$ is continuous since $i_x^{-1}(U \cup V) = U \in \mathcal{T}_X$. Similarly, i_y is continuous. Both i_x, i_y are open maps.

Suppose $f_x : X \rightarrow Z, f_y : Y \rightarrow Z$ are continuous maps to an auxiliary space Z . Then there exists a unique continuous $\phi : X \sqcup Y \rightarrow Z$ such that $\phi \circ i_x = f_x, \phi \circ i_y = f_y$. First, define $\phi : X \sqcup Y \rightarrow Z$ by

$$\phi(w) = \begin{cases} f_x(w) & \text{if } w \in X, \\ f_y(w) & \text{if } w \in Y. \end{cases}$$

Since X, Y are disjoint, ϕ is well-defined. Suppose $V \subset Z$ is open. Then $\phi^{-1}(V) = (\phi|_X)^{-1}(V) \cup (\phi|_Y)^{-1}(V) = f_x^{-1}(V) \cup f_y^{-1}(V)$ is open in $X \sqcup Y$, so ϕ is continuous.

Note that $\phi i_x(x) = \phi(x) = f_x(x) \rightarrow \phi i_x = f_x$. Similarly, $\phi i_y = f_y$.

Suppose there exists a map $\phi' : X \sqcup Y \rightarrow Z$ satisfying the same properties as ϕ . For $w \in X$, we have $\phi(w) = \phi(i_x(w)) = f_x(w) = \phi'(i_x(w)) = \phi'(w)$. Similarly, for $w \in Y$, $\phi(w) = \phi'(w)$, so $\phi = \phi'$ for all $w \in X \sqcup Y$.

Suppose $\mathcal{T}'$ is a topology on $X \sqcup Y$ such that i_x, i_y are continuous and the extension property holds. Let $W \in \mathcal{T}'$. Then $i_x^{-1}(W) = W \cap X \in \mathcal{T}_X$, $i_y^{-1}(W) = W \cap Y \in \mathcal{T}_Y$. Further, $W \cap X, W \cap Y$ have union W , so $\mathcal{T}' \subset \mathcal{T}$.

Now take $Z = X \sqcup Y$ with the topology $\mathcal{T}$. Let j_x, j_y be the (continuous) inclusions of X, Y into $X \sqcup Y$, respectively. Let $\phi : X \sqcup Y \rightarrow X \sqcup Y$ be the continuous function satisfying the extension property. If $w \in X$, then

$\phi(w) = \phi(i_x(w)) = j_x(w) = w$. Similarly, if $w \in Y$, $\phi(w) = w$. Let $U \cup V \in \mathcal{T}$. Then $\phi^{-1}(U \cup V) = U \cup V \in \mathcal{T}'$, so $\mathcal{T} \subset \mathcal{T}'$. Hence, $\mathcal{T} = \mathcal{T}'$.

Independence of $X \sqcup Y$. Suppose X, Y are topological spaces. Let X_1, X_2, Y_1, Y_2 be spaces such that $X_1, X_2 \cong X$ and $Y_1, Y_2 \cong Y$ such that $\{X_1, Y_1\}, \{X_2, Y_2\}$ are disjoint pairs. Then $X_1 \sqcup Y_1 \cong X_2 \sqcup Y_2$.

Let $h_1 : X_1 \rightarrow X, h_2 : X_2 \rightarrow X, k_1 : Y_1 \rightarrow Y, k_2 : Y_2 \rightarrow Y$ be homeomorphisms. Define $h : X_1 \sqcup Y_1 \rightarrow X_2 \sqcup Y_2$ by $h(w) = \begin{cases} h_2^{-1}h_1(w) & \text{if } w \in X_1, \\ k_2^{-1}k_1(w) & \text{if } w \in Y_1. \end{cases}$. Since $h_2^{-1}h_1, k_2^{-1}k_1$ are homeomorphisms, so is h .

5 Fundamental Group of Wedge of Circles

Claim: $G = \langle x_1, x_2, \dots, x_{n+1} \mid x_n x_{n+1}^{-1} \rangle \cong \langle y_1, y_2, \dots, y_n \rangle$.

Since $F_{n+1} = \langle x_1, x_2, \dots, x_{n+1} \rangle$ is free, by the extension property, let $\varphi : F_{n+1} \rightarrow F_n$ be the unique homomorphism to $F_n = \langle y_1, y_2, \dots, y_n \rangle$ such that $\varphi(x_i) = y_i$ for $i \in [n]$ and $\varphi(x_{n+1}) = y_n$. Since each generator of F_n is in the range of φ , φ is surjective.

Note that $\varphi(x_n x_{n+1}^{-1}) = y_n y_n^{-1} = 1$, so $M = N(\{x_n x_{n+1}^{-1}\}) < \ker \varphi$, so φ descends to an epimorphism $\bar{\varphi} : G \rightarrow F_n$.

Let $\phi : F_n \rightarrow F_{n+1}/M$ be given by $\phi(y_i) = x_i M$, well-defined by extension property. I claim ϕ is a left inverse of $\bar{\varphi}$, which implies $\bar{\varphi}$ is an isomorphism. It suffices to show that $\phi \bar{\varphi}$ fixes the generators of F_{n+1}/M .

For $i \in [n]$, $\phi \bar{\varphi}(x_i M) = \phi \varphi(x_i) = \phi(y_i) = x_i M$. For $i = n+1$, $\phi \bar{\varphi}(x_{n+1} M) = \phi \varphi(x_{n+1}) = \phi(y_n) = x_n M = M x_n = M x_n x_{n+1}^{-1} x_{n+1} = M x_{n+1} = x_{n+1} M$.

Let X be Hausdorff such that $X = \bigcup_{i \in [n]} X_i$ where $X_i \cong S^1$ and $X_i \cap X_j = \{p\}$ for some fixed point p if $i \neq j$. Let X' be defined similarly.

Consider the arbitrary pairing $X_i \rightarrow X'_i$. Let $h_i : X_i \rightarrow X'_i$ be a homeomorphism such that $p \mapsto p'$. Since S^1 is compact, so is X_i . Since X is Hausdorff and X_i is compact, X_i is closed. Extend by Gluing lemma $h : X \rightarrow X'$.

Suppose $h(x) = h(y)$. If $h(x), h(y) \in X'_i$ for some i , then $h_i(x) = h_i(y) \rightarrow x = y$, since h_i is injective. If $h(x), h(y)$ are in different circles, then $h(x) = h(y) = p' \rightarrow x = y = p$, so h is injective. Since each h_i is surjective, so is h , so h is bijective continuous.

Since $X = \bigcup X_i$ is a union of compact sets, X is compact. Then h is a continuous bijection from a compact space to a Hausdorff space, and therefore a homeomorphism.

Take X to be the embedding in the plane with a pencil of circles.

Let X be the wedge of circles $\{X_\alpha\}_{\alpha \in J}$ with center p . Let $f_\alpha : I \rightarrow X_\alpha$ be a generating loop of X_α . Let $\mathcal{K}$ be the set of all finite nonempty subsets of J . For $K \in \mathcal{K}$, let $X_K = \bigcup_{\alpha \in K} X_\alpha$. Let $i_\alpha : X_\alpha \rightarrow X$, $i_{\alpha,K} : X_\alpha \rightarrow X_K$, $i_K : X_K \rightarrow X$ be inclusions. Then $\pi_1(X, p) = \langle i_\alpha f_\alpha : \alpha \in J \rangle$.

Let H be an arbitrary group and $\{y_\alpha\}_{\alpha \in J} \subset H$. By the previous result, $\pi_1(X_K, p) = \langle i_{\alpha,K} f_\alpha : \alpha \in K \rangle$ is free. Let $h_K : \pi_1(X_K, p) \rightarrow H$ be such that $h_K(i_{\alpha,K} f_\alpha) = y_\alpha$ by extension property. Define $\rho : \mathcal{L}(X, p) \rightarrow H$ by picking $K \in \mathcal{K}$ such that $g(I) \in X_K$, letting g' be the restriction of g to the range X_K , and setting $\rho(g) = h_K([g'])$.

Suppose $K \subset K'$. Then I claim $h_K([g]) = h_{K'}([i_K^{K'} g])$ for all $g \in \pi_1(X_K, p)$. Let $\alpha \in K$, so $i_{\alpha,K} f_\alpha$ is a generator of $\pi_1(X_K, p)$. Then

$$h_{K'}([i_K^{K'} i_{\alpha,K} f_\alpha]) = h_{K'}(i_{\alpha,K'} f_\alpha) = y_\alpha = h_K(i_{\alpha,K} f_\alpha),$$

so h_K matches $h'_K i_{K*}^{K'}$ over $\pi_1(X_K, p)$.

Now suppose $g \in \mathcal{L}(X, p)$. Pick K, K' such that $g(I) \subset X_K, X_{K'}$. Then $g(I) \subset X_{K \cup K'}$, where $K \cup K' \in \mathcal{K}$. Let g', g'' be the restrictions. Then $h_K([g']) = h_{K \cup K'}([i_K^{K \cup K'} g']) = h_{K \cup K'}([i_{K'}^{K \cup K'} g'']) = h_{K'}([g''])$, since $i_K^{K \cup K'} g' = i_{K'}^{K \cup K'} g''$, so ρ is independent of the choice of K and therefore well-defined.

I claim $\rho(f * g) = \rho(f)\rho(g)$. Pick K such that $f(I), g(I) \subset X_K$ and let f', g' be the restrictions. Then $\rho(f * g) = h_K([(f * g)']) = h_K([f' * g']) = h_K([f'] [g']) = h_K([f']) h_K([g']) = \rho(f)\rho(g)$. Hence, ρ preserves multiplication.

Suppose $f \xrightarrow{H} g$ in X . Since I^2 is compact, so is $H(I^2)$. Let $H' : I^2 \rightarrow X_K$ be a restriction. Let f', g' be the corresponding induced restrictions $(H(\cdot, 0), H(\cdot, 1))$. Then $\rho(f) = h_K([f']) = h_K([g']) = \rho(g)$. Hence, ρ descends to a map $h : \pi_1(X, p) \rightarrow H$. For $f, g \in \mathcal{L}(X, p)$, we have $h([fg]) = \rho(fg) = \rho(f)\rho(g) = h([f])h([g])$, so h is a homomorphism. Further, $h([i_\alpha f_\alpha]) = \rho(i_\alpha f_\alpha) = h_{\{\alpha\}}(f_\alpha) = y_\alpha$, so $\pi_1(X, p) = \langle i_\alpha f_\alpha : \alpha \in J \rangle$ by Extension property.

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(a). Suppose FTSOC $r : B^{n+1} \rightarrow S^n$ is a retract. Since B^{n+1} is contractible, that is $\text{id}_{B^{n+1}}$ is null-homotopic, $r = r \circ \text{id}_{B^{n+1}}$ is null-homotopic. Then $\text{id}_{S^n} = \text{id}_{B^{n+1}}|_{S^n}$ is null-homotopic, contradiction since id_{S^n} is continuous an-

tipode preserving.

(b). Let X denote the upper hemisphere of S^{n+1} : $X = S^{n+1} \cap (\mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0})$. Let $p : X \rightarrow B^{n+1}$ be the geometric projection of X onto the $n+1$ -disc. Consider $f = g \circ p^{-1} : B^{n+1} \rightarrow S^n$. Since B^{n+1} is contractible, f is null-homotopic, so $f|_{S^n}$ is also null-homotopic. However, $f|_{S^n}$ is continuous antipode preserving, contradiction.

(c). Suppose FTSOC $f(x) \neq f(-x)$ for all $x \in S^{n+1}$. Define $g : S^{n+1} \rightarrow S^n$ by $g(x) = (f(x) - f(-x)) / \|f(x) - f(-x)\|$. Observe that g is continuous antipode preserving, contradicting (b).

Show that if X is an infinite wedge of circles S_n then X is not first-countable.

Let the circles have the common point p . Suppose FTSOC p has a countable neighborhood basis $\mathcal{N} = \{N_1, N_2, N_3, \dots\}$. Consider $N_n \cap S_n$, which is open in S_n . Since S_n has no isolated points and $N_n \cap S_n \neq \emptyset$, we can pick $x_n \in N_n \cap S_n$. Since S_n is T1, pick $U_n(p)$ not containing x_n . Let $U = \bigcup_{1 \leq n < \omega} U_n$.

Observe that since X is coherent with the subspaces S_n , U is open as $U \cap S_n = U_n$, so U is a neighborhood of p . By assumption there exists $n \in \mathbb{N}$ such that $N_n \subset U$. Then $N_n \cap S_n \subset U \cap S_n = U_n$, so $x_n \in N_n \cap S_n \subset U_n$, so $x_n \in U_n$, contradiction.

6 Adjoining Two-Cell

Let X be Hausdorff and $A \subset X$ be closed path connected. Suppose continuous $h : D \rightarrow X$ is such that $h(S^1) \subset A$ and h maps D° bijectively with $X - A$. Fix $p \in S^1$ and let $h(p) = a$. Then the homomorphism induced by inclusion $i_* : \pi_1(A, a) \rightarrow \pi_1(X, a)$ is surjective.

I claim $h|_{D^\circ}$ is a homeomorphism of D° with $X - A$. First, for any $B \subset D$, we have $h(B \cap D^\circ) = h(B) \cap h(D^\circ)$. It is always true for any function that $h(B \cap D^\circ) \subset h(B) \cap h(D^\circ)$. Suppose $y \in h(B) \cap h(D^\circ)$. Then $y \in h(D^\circ) = X - A$. Since h maps S^1 to A and $h|_{D^\circ}$ is bijective, there exists unique $x \in D^\circ$ such that $y = h(x)$. It is given $y \in h(B)$, so $x \in B$. It follows $y \in h(B \cap D^\circ)$.

Since D is compact and X is Hausdorff, h is a closed map. I claim that $h|_{D^\circ}$ is a closed map. Let $B \cap D^\circ \subset D^\circ$ be closed. Then $h(B \cap D^\circ) = h(B) \cap h(D^\circ) = h(B) \cap (X - A)$, which is closed in $X - A$ since $h(B)$ is closed. Since $h|_{D^\circ}$ is bijective continuous closed, $h|_{D^\circ}$ is a homeomorphism.

7 Absolute Retracts and Universal Extension Property

Let a space Y have the universal extension property if for all triplets (X, A, f) where X is a normal space, $A \subset X$ is closed, and $f : A \rightarrow Y$ is continuous, there exists an extension $g : X \rightarrow Y$.

Y is path connected. Suppose $y_0, y_1 \in Y$. Let $X = I$, $A = \{0, 1\}$ and $f : \{0, 1\} \rightarrow Y$ be such that $f(0) = y_0, f(1) = y_1$. Extend f to $g : I \rightarrow Y$, so g is a path between y_0, y_1 .

Y need not be Hausdorff. Take Y to have the trivial topology. Suppose $f : A \rightarrow Y$ is any function. Then f is continuous. Take $g : X \rightarrow Y$ to be any extension of f . g is also continuous.

Y is simply connected. Let $y_0 \in Y$. Suppose $f : I \rightarrow Y$ is a loop through y_0 . Let $X = I^2$, $A = \text{Bd } I^2$ and $g : A \rightarrow Y$ be such that $g(s, 0) = f(s), g(0, t) = g(1, t) = g(s, 1) = y_0$. Then g is continuous by gluing lemma. By extension property, let $H : I^2 \rightarrow Y$ extend g , so H is a path homotopy of f with e_{y_0} .

I can't use SDR of Y onto a point y_0 because $Y \times I$ need not be normal.

A convex subset of $\mathbb{R}^n$ need not have the universal extension property: let $Y = S^1$, $X = B^2$, $A = S^1$, $f = \text{id}_{S^1}$. Since f is antipode preserving continuous, f is not null homotopic. However, if we can extend f to $g : B^2 \rightarrow S^1$, then f is null homotopic, contradiction.

The product of spaces Y_α with the universal extension property has the universal extension property. Let $Y = \prod Y_\alpha$. Suppose $A \subset X$ is a closed subspace of the normal space X . Suppose $f : A \rightarrow Y$ is continuous. Then $f_\alpha = \pi_\alpha f : A \rightarrow Y_\alpha$ is continuous for all $\alpha \in J$. Let $g_\alpha : X \rightarrow Y_\alpha$ be an extension of f_α , $g = (g_\alpha)_{\alpha \in J} : X \rightarrow Y$.

For $x \in A$, $g(x) = (g_\alpha(x))_{\alpha \in J} = (f_\alpha(x))_{\alpha \in J} = f(x)$, so g extends f .

Suppose Y has the universal extension property and $Y' \xrightarrow{h} Y$. Then Y' has the universal extension property. Let $A \subset X$, $f' : A \rightarrow Y'$. Then $hf' : A \rightarrow Y$ is continuous, so extend to $g : X \rightarrow Y$. Let $g' : X \rightarrow Y'$ be $g' = h^{-1}g$.

I claim that g' extends f' . For $x \in A$, $g'(x) = h^{-1}(hf'(x)) = f'(x)$, as desired. Corollary, $(B^2)^\circ \cong \mathbb{R}^2$ has the UEP.

Let the normal space Y be called an absolute retract if for all pairs of spaces (X_0, Y_0) such that X_0 is normal, Y_0 is a closed subspace of X_0 and $Y_0 \cong Y$ that there exists a retract $r : X_0 \rightarrow X_0$ from X_0 onto Y_0 .

Embed $Y \hookrightarrow I^J$ for some set J (since Y is completely regular). Let $Y_0 \cong Y$ be the homeomorphic subspace of I^J . Then there exists a retract $r : I^J \rightarrow Y_0$.

Suppose $Y \cong Y'$, where Y is an absolute retract. Then Y' is an absolute retract. Let (X_0, Y'_0) be such that X_0 is normal, $Y'_0 \subset X_0$, and $Y'_0 \cong Y'$. Then $Y'_0 \cong Y$, so there exists a retract $r : X_0 \rightarrow X_0$ from X_0 onto Y'_0 .

Suppose Y is normal and has the UEP. Then Y is an absolute retract. Let (X_0, Y_0) . Let $h : Y_0 \rightarrow Y$ be the homeomorphism. Since $Y_0 \subset X_0$ is closed, extend to $f : X_0 \rightarrow Y$. Let $g = h^{-1}f : X_0 \rightarrow Y_0$. For $x \in Y_0$, we have $g(x) = h^{-1}f(x) = h^{-1}h(x) = x$, so g (upon composition with inclusion) is a retract.

Suppose Y is normal with UEP and $Z \subset Y$ is a retract. Then Z has the UEP. Let (X, A, f) be as usual. Let $i : Z \rightarrow Y$ be inclusion. By extension property of Y , let $g : X \rightarrow Y$ extend if . Let $h : X \rightarrow Z$ be rg upon restriction. For $x \in A$, $h(x) = rg(x) = rif(x) = f(x)$.

8 Adjunction Space

Let $f : A \rightarrow X$ be continuous, $i_X : X \rightarrow X \sqcup Y, i_Y : Y \rightarrow X \sqcup Y$ be inclusions, $p : X \sqcup Y \rightarrow X \cup_f Y$ be the quotient map. Let $\phi_X : X \rightarrow X \cup_f Y$ be $\phi_X = pi_X$. Define similarly $\phi_Y : Y \rightarrow X \cup_f Y$. Claim: ϕ_X is a closed embedding.

Proof: It suffices to show ϕ_X is a closed injective map. Suppose $pi_X(x) = pi_X(x')$. Then $p^{-1}(pi_X(x)) = p^{-1}(pi_X(x'))$, so $i_X(x) \cup i_Y(f^{-1}(x)) = i_X(x') \cup i_Y(f^{-1}(x')) \rightarrow i_X(x) = i_X(x')$ (intersecting with $i_X(X)$). Since i_X is injective, $x = x'$.